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Dr.D.Y.PatilCollegeofEngineering&Innovation
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SMARTBRIDGE | SKILLWALLET

FINAI: AI-Powered Loan Eligibility Checker

Project Description :-

FINAI is an AI-powered loan eligibility and financial planning platform built for the Banking, Financial Services and Insurance (BFSI) domain, designed to help borrowers understand their indicative loan eligibility within the Indian financial ecosystem. The platform evaluates key applicant criteria such as monthly salary, age, credit score (CIBIL benchmark), existing EMI obligations and loan tenure against configurable underwriting rules.

The application provides an interactive dashboard with a Chart.js doughnut chart for principal versus interest, a credit score gauge meter, a factor rating radar chart and a searchable application history. Every evaluation is stored as an audit trail so users can review past results, export them as CSV and sync them to Google Sheets.

FINAI is designed with a modern, responsive glassmorphism interface and uses HTML5, CSS3, JavaScript (ES6+), Node.js, Express.js, Chart.js, Anthropic Claude API and Google Sheets integration to create a production-oriented FinTech web application, deployed live on Vercel.

Technology & AI :-

Frontend: HTML5 / CSS3 (Glassmorphism Design System) / Vanilla ES6+ JavaScript / Chart.js / Lucide Icons

Backend: Node.js / Express.js

Data Storage: Google Sheets API v4 / Google Apps Script / Local JSON Persistence

AI: Anthropic Claude API (Claude 3.5 Sonnet) / Built-in BFSI Domain Knowledge Engine

API: RESTful APIs using Express.js

Authentication: bcryptjs Password Hashing / JSON Web Tokens (JWT) / Protected Routes

Security: Helmet (CSP) / CORS / Express Rate Limiter / Environment Variables for API Keys

Testing: Automated unit tests for business rules and financial calculations (npm test)

Deployment: Vercel (https://finai-opal.vercel.app)

Version Control: Git / GitHub

Key Features :-

• AI-powered loan eligibility checking

• Underwriting rules: minimum salary Rs. 30,000, minimum CIBIL score 700, existing EMI up to Rs. 20,000, age 21 to 65 years

• Indicative borrowing limit calculation (Monthly Salary × 20)

• Pass/Fail criteria breakdown with actionable improvement suggestions

• Claude AI financial assistant with context-aware chat

• BFSI Domain Knowledge Engine fallback when no API key is configured

• Application history with search, filters and detail view

• Google Sheets sync via Apps Script webhook or Sheets API v4

• Secure registration, login and protected routes

• Rate limiting, input sanitization and Content Security Policy

• Responsive dashboard with mobile drawer navigation

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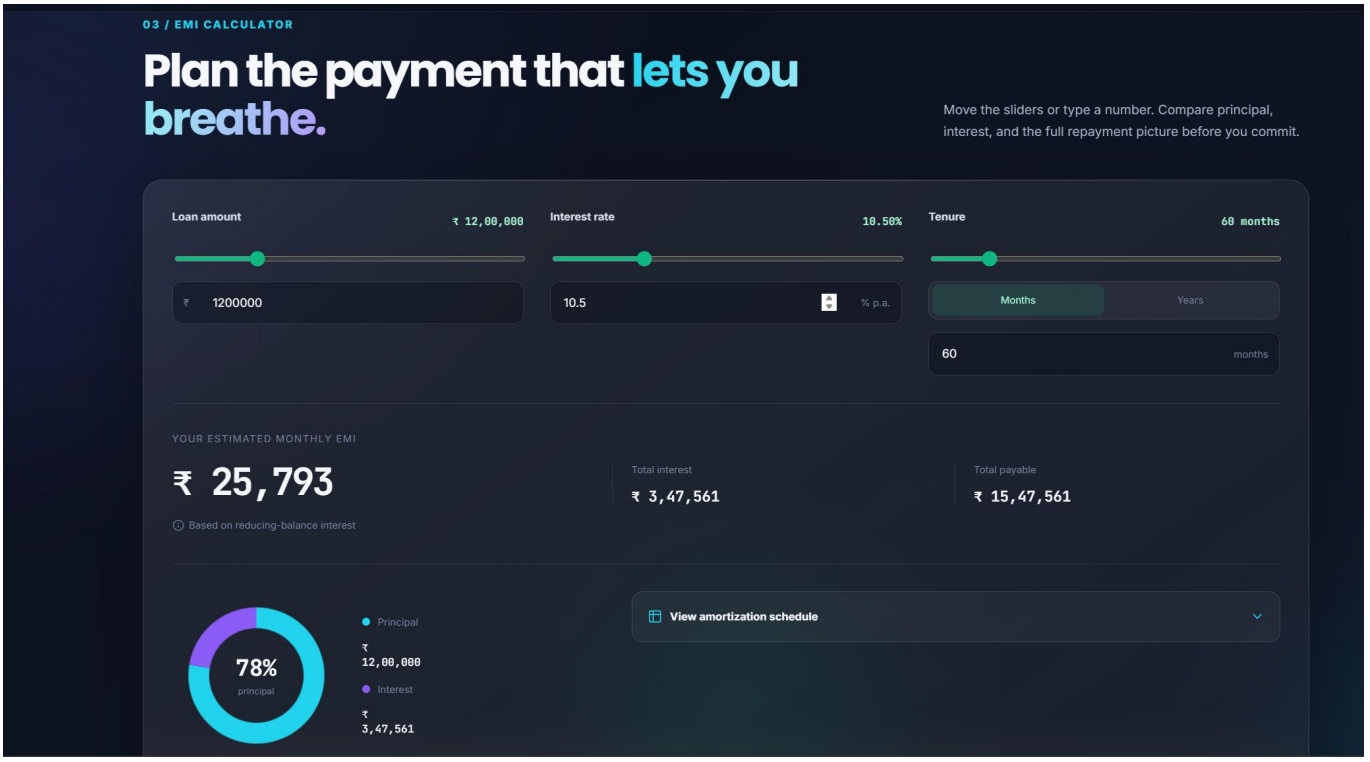


Scenarios :-

Scenario 1: A salaried applicant, Rahul Sharma (age 23, monthly salary Rs. 50,000, CIBIL score 560, existing EMI Rs. 12,00), requests a loan of Rs. 2,00,000 for 17 months. The system validates the inputs, checks every underwriting rule, and displays "Highly Eligible" with an indicative limit of Rs. 11,00,000 (Rs. 50,000 × 22), a debt-to-income ratio of 14% and personalized tips.

Scenario 2: A user with a "Fair" credit profile opens the Credit Score Analyzer and enters payment history, credit utilization, history length, active loans and recent inquiries. FINAI displays the score gauge, a five-factor radar chart and improvement strategies, and the user then asks the Claude AI assistant how to increase their loan limit and receives plain-language guidance.

Scenario 3: A borrower plans a loan of Rs. 14,00,000 at 7.5% annual interest for 35 months. The EMI calculator computes the monthly installment, total interest and total repayment, updates the doughnut chart instantly as the sliders move, and generates a full amortization schedule that can be exported to CSV.



Scenario 4: A user returns to the dashboard after several evaluations. The Application History page lists all saved records with reference IDs, allows search by name or status, shows full details in a modal, exports data to CSV and syncs each record to Google Sheets, falling back to local JSON storage if Sheets is not configured.

Milestones :-

Milestone 1 – Project Planning & Requirement Analysis

Define the objectives of FINAI, identify target users, finalize eligibility rules and features, and design the application workflow and technical architecture.

Milestone 2 – Environment Setup & Backend Development

Set up the Node.js and Express.js backend and static frontend structure, create the configuration module with business rules, and implement secure environment-variable management.

Milestone 3 – Authentication & Security Layer

Develop user registration and login with bcryptjs hashing and JWT sessions, protected routes, input validation and sanitization, rate limiting, Helmet CSP and safe error handling.

Milestone 4 – Eligibility, EMI & Credit Analytics

Implement the eligibility engine, reducing-balance EMI calculation with amortization schedules, credit score analysis across five weighted factors (35%, 30%, 15%, 10%, 10%) and application record management.

Milestone 5 – AI Assistant, Dashboard & Integrations

Integrate Anthropic Claude with the BFSI Domain Knowledge Engine fallback, build the interactive dashboard with Chart.js visualizations, application history, CSV export and Google Sheets synchronization.

Milestone 6 – Testing, Optimization & Deployment

Run the automated test suite (12 tests covering business rules, EMI mathematics and credit analysis), fix errors, improve performance and responsiveness, and deploy the application live on Vercel.

Conclusion :-

FINAI provides an intelligent and interactive way for users to understand their loan eligibility and financial health. By combining rule-based underwriting logic, EMI simulation, credit score analysis, data visualization and Generative AI, the platform transforms complex lending criteria into clear and actionable guidance.

The project demonstrates how modern technologies such as Node.js, Express.js, Chart.js, Google Sheets integration and Anthropic Claude can be combined to develop an AI-powered FinTech solution for loan pre-qualification, credit awareness and financial planning.

Expected Outcome :-

FINAI aims to help users better understand their eligibility and credit profile and make more informed borrowing decisions. The platform provides automated eligibility checks, EMI calculations, credit score analysis, personalized AI explanations and a complete application history in a single application.

The project will demonstrate practical expertise in Generative AI, FinTech, Node.js, Express.js, REST APIs, JavaScript, data visualization, authentication and security, Google Sheets integration, automated testing, prompt-driven AI assistance and cloud deployment, while providing a scalable foundation for future integration with real-world credit bureau and banking data.